

Software Setup & Environment

Trainee Onboarding — Windows Verification Guide

Every trainee independently installs, configures, verifies, clones, builds, runs, tests, commits, and pushes before assigned project work. Day 1 follows the hour plan. Mentors use the 10-minute station check.

DOCUMENT TYPE

Trainee Onboarding Standard

PRIMARY STACK

Java · Spring Boot · React · MySQL · Git · Docker

ALTERNATE BACKEND

Python / FastAPI when assigned

PLATFORM

Windows 10 / 11

MOBILE TRACK

Flutter + Android Studio

AUDIENCE

Trainees, mentors, coordinators, admin

Install. Configure. Verify. Then build.

INTERNAL TALENT DEVELOPMENT STANDARD

DOCUMENT	VERSION	CLASSIFICATION	OWNER
DTTP Software Setup	1.2	Internal Onboarding Standard	Deshmukh Technologies

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PART I Purpose & machine

01 Purpose

This document is the standard software and development environment for every Deshmukh Technologies trainee. Day 1 is not a lecture day. It is the day the machine becomes a working engineering station. Read this guide **before** you write Employee Management code.

INSTALL

CONFIGURE

VERIFY

CLONE

BUILD

RUN

TEST

COMMIT

PUSH

Document	Role
This setup guide	Machine, tools, GitHub, Day-1 sign-off
90-Day DTTP Handbook	Timed training path, project, assessment
Self-Growth Program	Ten-level knowledge map

Complete this path before assigned project development. Employee Management features do not start until the mentor marks the environment Pass or Conditional.

02 Scope

Installation, Windows layout, Git, GitHub SSH, Java / Spring Boot, Node / React, MySQL, Postman, Docker with WSL 2, Flutter for the mobile track, verification, troubleshooting, Day-1 hour plan, official product names, trainee sign-off, mentor 10-minute station check, and Day-1 completion. Python / FastAPI is installed only when that backend track is assigned. JDK remains the DTTP default.

03 Standard software checklist

Software	Purpose	Required
Windows 10 / 11	Operating system	Yes
Git + GitHub account	Version control and source	Yes
VS Code	General / frontend	Yes
IntelliJ IDEA	Java / Spring Boot	Yes
JDK 17 (or 21 if the project says so)	Java	Yes
Maven	Java build	Yes
Node.js LTS + npm	React	Yes
MySQL Server + Workbench	Database	Yes
Postman + Chrome	API and debugging	Yes
Docker Desktop	Containers	Recommended
Flutter SDK + Android Studio	Mobile	Mobile track
Python 3.11+	FastAPI	Python track only

04 Hardware requirements

MINIMUM

- 8 GB RAM
- 256 GB SSD
- Modern dual / quad-core CPU
- Stable internet
- Windows 10 / 11

RECOMMENDED

- 16 GB RAM
- 512 GB SSD
- Intel Core i5 / AMD Ryzen 5 or better
- Full HD display
- Reliable broadband

A machine below the minimum will struggle with IntelliJ + Chrome + Docker + MySQL at once. Tell the mentor before Day 1 if the machine is below minimum.

05 Windows preparation

Update: Settings → Windows Update → Check for updates. Install security and system updates.

Restart after major updates. Do not install JDK, Docker, or Android Studio on a pending-reboot machine.

Create C:\Development\ with Projects, Tools, Documents, and Backups.

Do not keep clones in C:\Windows, C:\Program Files, Desktop, or Downloads.

Recommended: enable Windows long paths from an Administrator PowerShell so deep Node and Java trees do not fail.

```
New-ItemProperty -Path "HKLM:\SYSTEM\CurrentControlSet\Control\FileSystem" `
-Name "LongPathsEnabled" -Value 1 -PropertyType DWORD -Force
```

PART II Tools

06 Git installation

Install Git for Windows. Enable Git from the command line, bundled OpenSSH, default branch `main`. Open a **new** PowerShell:

```
git --version
git config --global user.name "Your Name"
git config --global user.email "your-email@example.com"
git config --global init.defaultBranch main
git config --global --list
```

Expected: `git version 2.x.x`. Use the email on the GitHub profile.

07 GitHub account

Each trainee has an authorized GitHub account and repository access from the mentor or admin. Never share passwords, personal access tokens, private SSH keys, or company secrets. Access is granted to named people, not shared logins.

08 GitHub SSH configuration

```
ssh-keygen -t ed25519 -C "your-email@example.com"
Get-Service ssh-agent | Set-Service -StartupType Automatic
Start-Service ssh-agent
ssh-add $env:USERPROFILE\.ssh\id_ed25519
Get-Content $env:USERPROFILE\.ssh\id_ed25519.pub
```

Accept the default key path. Set a passphrase. If `Set -Service` is denied, run those two service lines as Administrator, then `ssh-add` in a normal window.

Copy the **public** key only (starts with `ssh-ed25519`, one line). GitHub → Settings → SSH and GPG keys → New SSH key. Title: DTTP-Windows-<name>.

```
ssh -T git@github.com
Hi <github-username>! You've successfully authenticated, but GitHub does not provide shell access.
```

First connection may ask to trust GitHub's host key. Type yes. Never copy `id_ed25519` without `.pub`.

09 VS Code

Prettier

ESLint

GitLens

EditorConfig

Docker

REST Client

Java pack

Spring Boot pack

Format On Save: On. Auto Save: afterDelay or onFocusChange. Default formatter: Prettier for JavaScript, TypeScript, and JSON. React snippet packs are optional; prefer typing the code in training. Line endings stay **LF** even on Windows.

Command Palette (Ctrl+Shift+P) → **Preferences: Open User Settings (JSON)**. DTTP default for React + TypeScript:

```
{
  "editor.formatOnSave": true,
  "editor.defaultFormatter": "esbenp.prettier-vscode",
  "editor.tabSize": 2,
  "files.eol": "\n",
  "files.insertFinalNewline": true,
  "[typescript]": { "editor.defaultFormatter": "esbenp.prettier-vscode" },
  "[typescriptreact]": { "editor.defaultFormatter": "esbenp.prettier-vscode" }
}
```

If the frontend starter has no Prettier file, add `.prettierrc` with `singleQuote`, `semi`, `trailingComma: "es5"`, `printWidth: 100`.

10 IntelliJ IDEA

Use IntelliJ for Java, Spring Boot, Maven, JUnit, Mockito, JPA, and Spring Security. Community is enough unless the company issues Ultimate.

Setting	Value
Project SDK	JDK 17 (or 21 if the project says so)
Build tool	Maven
Version control	Git
Annotation processing	Enabled for Spring projects

Do not mix VS Code and IntelliJ on the same Java change in the same hour.

FIRST OPEN OF THE MAVEN PROJECT

File → Open → folder that contains `pom.xml` (not a parent zip).

Trust the project. Wait for Maven import; reload the Maven tool window if it hangs.

Project Structure (Ctrl+Alt+Shift+S) → Project SDK **17** → Language level 17.

Maven home = wrapper or Bundled. Runner JRE = 17. Enable annotation processing.

Run the `@SpringBootApplication` class. Then `curl http://localhost:8080/api/health`.

If IntelliJ says “SDK is not defined”: Project Structure → Project SDK → 17 → Apply → Maven reload.

11 Java JDK

DTTP default: **JDK 17 LTS**. Install a full JDK, not a JRE.

```
JAVA_HOME = C:\Program Files\Java\jdk-17
Path      += %JAVA_HOME%\bin
```

Point `JAVA_HOME` at the JDK root, not at `bin`. Folder names vary by vendor (Temurin, Microsoft, Oracle). New terminal:

```
java -version
javac -version
echo $env:JAVA_HOME
```

Both commands must work. `java` without `javac` means a JRE is on PATH. Expected: `openjdk version "17.0.x"` and `javac 17.0.x`.

SET JAVA_HOME IN THE WINDOWS GUI

Start → type **environment variables** → Edit the system environment variables

Environment Variables...

System variables → New... Name `JAVA_HOME`, value the JDK folder (not `bin`)

Edit Path → New → `%JAVA_HOME%\bin`

OK. Open a **new** PowerShell.

12 Maven

Install Maven 3.9.x or use the project wrapper. Prefer the wrapper when `mvnw.cmd` exists:

```
mvn -version
.\mvnw.cmd clean install
```

Do not upgrade Maven globally to fix a wrapper build.

13 Spring Boot verification

```
cd C:\Development\Projects
git clone git@github.com:<org>/<backend-repo>.git
cd <backend-repo>
.\mvnw.cmd clean install
.\mvnw.cmd spring-boot:run
```

Copy local config from the example. Never commit it.

```
spring.datasource.url=jdbc:mysql://localhost:3306/dtpp_training
spring.datasource.username=dtpp_dev
spring.datasource.password=use-a-local-password
```

Pass when: BUILD SUCCESS, process stays up, logs show MySQL connected, GET `http://localhost:8080/api/health` returns `{ "status": "ok" }` (or the project health path).

14 Node.js

Install the Node.js LTS that matches the frontend (`package.json` engines and `.nvmrc` if present). DTPP frontend is **React + TypeScript**. Expected: Node 20.x or 22.x unless the project pins another version.

```
node --version
npm --version
```

Do not change Node versions to “fix” `npm install` on an existing lockfile.

15 React environment

```
cd C:\Development\Projects
git clone git@github.com:<org>/<frontend-repo>.git
cd <frontend-repo>
copy .env.example .env
npm install
npm run dev
```

Some projects use `npm start`. Follow `package.json`. Chrome must load the printed URL (often `http://localhost:5173`). Be able to point at components, props, state, hooks, routing, service-layer API calls, forms, errors, and env vars. Never commit secrets.

16 MySQL installation

Install MySQL Server 8.x and Workbench. Local root password in a password manager, not in chat. Windows service enabled. Port **3306**.

Field	Value
Host	localhost
Port	3306
Training database	dttp_training

Never use production credentials on a trainee machine.

17 MySQL verification

The trainee must create a database, create a table, insert, query, update, and delete.

If the assigned repo uses **PostgreSQL** instead: port 5432, same database name `dttp_training`, JDBC URL `jdbc:postgresql://localhost:5432/dttp_training`. Do not mix MySQL and PostgreSQL drivers.

```
CREATE DATABASE IF NOT EXISTS dttp_training
  CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci;
CREATE USER IF NOT EXISTS 'dttp_dev'@'localhost' IDENTIFIED BY 'use-a-local-password';
GRANT ALL PRIVILEGES ON dttp_training.* TO 'dttp_dev'@'localhost';
USE dttp_training;
CREATE TABLE smoke_test (id INT PRIMARY KEY AUTO_INCREMENT, label VARCHAR(80) NOT NULL);
INSERT INTO smoke_test (label) VALUES ('DTTP Day 1');
SELECT * FROM smoke_test;
```

18 Postman

Learn GET, POST, PUT, PATCH, DELETE, URL, query, path, headers, JSON, Bearer token, and status codes.

[POST /API/AUTH/LOGIN](#)[JWT](#)[AUTHORIZATION BEARER](#)[GET /API/EMPLOYEES](#)[GET /API/HEALTH](#)

Save the collection under `C:\Development\Documents`. No production tokens.

19 Docker Desktop

WSL 2 backend is required. Complete WSL **before** you expect `docker run hello-world` to work. Kubernetes is not required for DTTP.

PowerShell as Administrator:

```
wsl --install
wsl --status
wsl --set-default-version 2
```

Restart when Windows asks. If virtualization is disabled: BIOS/UEFI → enable Intel VT-x or AMD-V / SVM. Company laptop: ask IT. Do not share BIOS passwords.

Docker Desktop → Settings → General → Use the WSL 2 based engine. Apply & Restart.

```
wsl --status
docker --version
docker compose version
docker run hello-world
```

Understand image, container, Dockerfile, volume, network, Compose. On 8 GB RAM, start Docker only when the day's work needs it. Mentor may waive Docker in writing.

20 Flutter — mobile track

Install Flutter SDK (stable), Android Studio, Android SDK, emulator or authorized device with USB debugging. Add Flutter bin to PATH.

```
flutter --version
flutter doctor
flutter doctor -v
cd C:\Development\Projects
flutter create dttp_test_app
cd dttp_test_app
flutter run
```

Pass when the counter app launches. Resolve every red `flutter doctor` item on this track.

21 Chrome DevTools

Press **F12**. Elements (HTML/CSS), Console (JS errors), Network (API status and payload), Application (storage and cookies), Sources (breakpoints). Day-1: load the React app and confirm the document request, plus one XHR/fetch when the API is up.

PART III Verify

22 Environment verification

Run in a **new** PowerShell. Paste output into the sign-off.

```
git --version
java -version
javac -version
mvn -version
node --version
npm --version
mysql --version
docker --version
docker compose version
ssh -T git@github.com
```

Mobile: flutter --version and flutter doctor. Python track: python --version (3.11 or 3.12) and pip --version. Skip Docker commands only with a written mentor waiver.

23 Project verification

01 GitHub repository	02 Clone with SSH
03 Install dependencies	04 Configure local env (not committed)
05 Database connection	06 Build
07 Run backend	08 Run frontend
09 Test API	10 Small allowed change
11 Commit	12 Push

First commit only after the mentor names a safe file:

```
git checkout -b dttp/<initials>/env-setup
git add docs/setup-notes.md
git commit -m "docs: record Day-1 environment verification"
git push -u origin dttp/<initials>/env-setup
```

Do not commit .env or application-local.properties.

SAMPLE DOCS/SETUP-NOTES.MD

```
# Day-1 environment
- OS: Windows 11
- RAM: 16 GB
- JDK: 17 (javac 17.0.x)
- Node: 22.x
- MySQL: 8.x on 3306, database dtp_training
- Docker: installed / waived
- GitHub SSH: pass
- Backend health: GET /api/health
- Frontend: npm run dev
- Secrets: none committed
```

24 Environment variables

Never hard-code `DB_PASSWORD`, `JWT_SECRET`, `API_KEY`, or AWS keys. Use `.env`, `application-local.properties`, or `application-dev.properties`. Confirm secrets are in `.gitignore`.

25 Recommended .gitignore

```
.env
.env.local
application-local.properties
.idea/
.vscode/
node_modules/
build/
dist/
target/
*.log
*.jks
*.pem
*.p12
id_ed25519
```

If a secret was committed by mistake, tell the mentor immediately. Do not only delete the file in a later commit.

26 Troubleshooting

Symptom	First check
Git / Java / Node not recognized	Install, PATH, new PowerShell. JDK needs <code>javac</code> , not only <code>java</code> .
Maven not recognized	Use <code>.\mvnw.cmd</code> in the project.
<code>npm install</code> fails	Match Node to <code>engines</code> / <code>.nvmrc</code> . Do not change lockfile versions at random.
MySQL connection fails	Service running, port 3306, user, password, database exists, host is <code>localhost</code> . PowerShell: <code>Get-Service *mysql*</code> then <code>Start-Service MySQL80</code> .
Port 8080 in use	<code>netstat -ano findstr :8080</code> — last column is PID. End task, or run on 8081 and record it in setup notes.
Docker does not start	Desktop running, virtualization, WSL 2 (<code>wsl --install</code> , <code>wsl --status</code>), <code>docker info</code> .
SSH asks for a GitHub password	Remote is HTTPS. Set origin to <code>git@github.com:<org>/<repo>.git</code> .
IntelliJ “SDK is not defined”	Project Structure → Project SDK 17 → Apply → Maven reload.
Flutter doctor red	<code>flutter doctor -v</code> . Fix SDK, licenses, emulator, or PATH first.

27 Security rules

☐ Never share passwords on WhatsApp or email	☐ Never upload company source to unauthorized locations
☐ Never commit credentials to GitHub	☐ Never use production credentials for training
☐ Never expose API keys	☐ Never install unauthorized software on a company machine
☐ Never share private SSH keys	☐ Report security incidents immediately

PART IV Sign-off

28 Trainee setup checklist

WINDOWS & TOOLS

- ☐ Windows updated and restarted
- ☐ C:\Development folders created
- ☐ Git identity configured
- ☐ VS Code + IntelliJ installed
- ☐ VS Code format-on-save + Prettier
- ☐ java and javac both work
- ☐ Maven wrapper or mvn verified
- ☐ Node LTS and npm verified
- ☐ Chrome installed

DATA, GITHUB, TRACKS

- ☐ MySQL 3306 + dtp_training smoke SQL
- ☐ Postman collection saved
- ☐ Docker hello-world or written waiver
- ☐ WSL 2 status OK if Docker required
- ☐ SSH public key on GitHub
- ☐ ssh -T git@github.com succeeded
- ☐ Assigned repo cloned with SSH
- ☐ Mobile: flutter doctor + test app
- ☐ Python track: python 3.11+ if assigned

29 Python / FastAPI track

Install only when the mentor assigns this track. Same MySQL database, Git, and security rules. Health remains GET /api/health.

```
python --version
pip --version
python -m venv .venv
.\\.venv\\Scripts\\Activate.ps1
pip install -r requirements.txt
uvicorn app.main:app --reload
```

If activation is blocked: Set-ExecutionPolicy -Scope CurrentUser RemoteSigned.

30 Evidence pack

Attach screenshots or pasted logs to the Day-1 report. Crop secrets.

Evidence	Shows
<code>git --version</code> and global config	Git identity
<code>ssh -T git@github.com</code>	SSH works
<code>java / javac</code> and <code>Node / npm</code>	JDK and Node
Workbench on <code>dtp_training</code>	Database
Backend BUILD SUCCESS or run log	Spring Boot
Browser with the React app	Frontend
Postman GET <code>/api/health</code>	API
GitHub setup branch	Push
<code>flutter doctor</code> (mobile)	Mobile track

31 Trainee sign-off

A verbal “it’s done” is not Day-1 complete.

Field	Trainee fills
Full name	
Date	
GitHub username	
Machine (make / RAM / disk)	
Windows version	
Backend track (Java / Python)	
Mobile track (Yes / No)	
Docker (Installed / Waived)	
Assigned repos	
I confirm that the required tools are installed, SSH works, I can clone, build, and run the assigned starter, no secrets are in Git, and the evidence pack is attached. Trainee signature / typed name _____ Date _____	

32 Mentor / admin verification

Spot-check: `ssh -T, java and javac (or Python), MySQL dtp_training, one running app, .gitignore covers .env, remote is git@github.com: not HTTPS`. Then run the **10-minute station check** (\$35) at the trainee machine. Do not sign Pass from screenshots of a different PC.

Result	Meaning
Pass	Day-1 complete. Trainee may start Phase 1 technical work.
Conditional	Listed gaps with a fix date within 48 hours.
Fail	Environment is not usable. Repeat setup with mentor support.

Mentor name

Result — Pass / Conditional / Fail

Notes

Date

SAMPLE CONDITIONAL NOTE

```
Result: CONDITIONAL
Gaps (fix by 20 Aug, 17:00):
  1. javac not on PATH – set JAVA_HOME, new terminal.
  2. Origin is HTTPS – set-url to git@github.com:org/repo.git
Docker waived (8 GB RAM) until Week 10.
```

33 Day-1 hour plan

Do **not** start Employee Management screens on Day 1. Finish this clock. If you fall behind, skip Flutter and Docker polish — finish Git, JDK, Node, MySQL, clone, health, first commit, and the sign-off table.

Hour	Block	Done when
0–1	Folders, Git identity, SSH, GitHub test	<code>ssh -T git@github.com</code> succeeds. Company email in Git config.
1–2	JDK 17, JAVA_HOME, Maven wrapper, IntelliJ	<code>java -version</code> is 17. <code>.\mvnw.cmd -v</code> works. SDK is 17.
2–3	Node LTS, VS Code + ESLint + Prettier	Node 20 or 22. Format-on-save in settings.json.
3–4	MySQL 8, dtp_training, Postman	Smoke SQL ran. Postman echo GET is 200.
4–5	Clone, /api/health, first commit, sign-off	Health JSON. GitHub shows the commit. Mentor has the table.

If the starter repo is late: still finish tools and `dtp_training`. Write that in `docs/setup-notes.md`. Do not invent a private Spring project as a substitute.

Day 1 is not EMS leave APIs, AWS, Kubernetes, Flutter (unless assigned), or production Docker images.

34 Official product names

Download from the vendor. Do not use a random “JDK 17” zip from a blog.

Tool	What to download
Git	Git for Windows — git-scm.com
JDK 17	Eclipse Temurin 17 (x64 MSI) or Microsoft Build of OpenJDK 17
IntelliJ	IntelliJ IDEA Community Edition
Node	Node.js LTS Windows Installer (.msi) x64
VS Code	Visual Studio Code User Installer x64
MySQL	MySQL Installer for Windows — Server 8.0
PostgreSQL	EnterpriseDB Windows installer — only if assigned
Postman	Postman for Windows
Docker	Docker Desktop for Windows
Flutter	Flutter SDK Windows zip — mobile track only
Python	Python 3.12 from python.org — FastAPI track only

35 Mentor 10-minute station check

Sit at the trainee machine. Run these commands. Do not accept screenshots of a different PC.

```
git --version
git config --global user.name
git config --global user.email
ssh -T git@github.com
java -version
echo $env:JAVA_HOME
node -v
npm -v
Get-Service *mysql*
```

cd into the cloned starter. `git remote -v` must be GitHub, not a USB copy.

`.\mvnw.cmd -q -DskipTests spring-boot:run` (or IntelliJ Run). Wait until the process starts.

`curl http://localhost:8080/api/health` — JSON with a healthy status.

`mysql -u dtp_dev -p -e "USE dtp_training; SELECT 1;"` — password from local notes, not chat.

Open committed `docs/setup-notes.md` — not a file that exists only on Desktop.

GitHub.com → assigned repo → latest allowed commit author is the trainee.

Pass in 10 minutes: identity, SSH, JDK 17, Node, MySQL service running, health JSON, notes file on GitHub.

Fail immediately if user.email is personal Gmail, JAVA_HOME is empty, health never returns, MySQL is Stopped and the trainee cannot start it, or the repo is missing.

36 Day-1 completion criteria

- | | |
|--|---|
| ☐ Core verification commands succeed | ☐ Backend or frontend starter runs locally |
| ☐ GitHub SSH authentication succeeds | ☐ GET /api/health returned JSON |
| ☐ Git identity uses the company email | ☐ One API call shown in Postman or Network |
| ☐ JDK 17 on PATH and JAVA_HOME set (new terminal) | ☐ One allowed commit pushed with docs/setup-notes.md |
| ☐ Node LTS on PATH; VS Code format-on-save on | ☐ Trainee sign-off filled |
| ☐ Assigned repo cloned into C:\Development\Projects | ☐ Mentor result after the 10-minute station check |
| ☐ MySQL (or PostgreSQL) service Running; dttp_training exists | ☐ Docker is Pass or Conditional with a dated next action |

Until Day 1 is complete, the trainee does not start Employee Management features. A broken machine produces fake velocity.

DTTP — DESHMUKH TECHNOLOGIES TRAINEE PROGRAM

Install. Configure. Verify. Then build.

Windows station ready. GitHub authenticated. Database local. First commit on a branch.

Learn. Build. Solve. Review. Deploy. Grow.